

BASE NANOPROCESSOR - MACHINE CODE & OPCODES GUIDE

CS1050 - Computer Organisation & Digital Design
Team WCSS - 12-bit Basic NanoProcessor

1. INSTRUCTION FORMAT (12-bit)

Bit position: 11 10 9 8 7 6 5 4 3 2 1 0

Opcode (2-bit) | R1 Dest (3-bit) | R2 Src (3-bit) | Immediate/Jump (4-bit)

2. OPCODE TABLE

Opcode Bits	Instruction	Operation	Format
10	MOVI	R = immediate	10 RRR 000 dddd
00	ADD	Ra = Ra + Rb	00 RaRaRa RbRbRb 0000
01	NEG	R = -R	01 RRR 0000 0000
11	JZR	If R=0 then PC = d	11 RRR 0000 0ddd

3. REGISTER ADDRESSES (RRR)

Binary	Register	Special Function
000	R0	Hardwired to 0 (read-only)
001-110	R1-R6	General purpose
111	R7	Displayed on 7-segment & LEDs

4. INSTRUCTION BREAKDOWN

4.1 MOVI (Move Immediate) - Opcode 10

Format: 10 RRR 000 dddd

What it does: Loads a constant value directly into a register.

Example: MOVI R7, 5 -> Machine Code: 0xB85

4.2 ADD (Add Two Registers) - Opcode 00

Format: 00 RaRaRa RbRbRb 0000

What it does: Adds two registers, stores result in Ra.

Example: ADD R7, R1 -> Machine Code: 0x390

4.3 NEG (Negate) - Opcode 01

Format: 01 RRR 0000 0000

What it does: Negates a register ($R = -R$) using 2's complement.

Example: NEG R7 -> Machine Code: 0x780

4.4 JZR (Jump if Zero Register) - Opcode 11

Format: 11 RRR 0000 0ddd

What it does: If register is zero, jump to address d (0-7).

Example: JZR R0, 4 -> Machine Code: 0xC04

5. DEMO PROGRAM MACHINE CODE

Addr	Instruction	Binary	Hex	Operation
0	MOVI R7, 0	1011 1000 0000	0xB80	R7 = 0
1	MOVI R1, 1	1000 1000 0001	0x881	R1 = 1
2	MOVI R2, 2	1001 0000 0010	0x902	R2 = 2
3	MOVI R3, 3	1001 1000 0011	0x983	R3 = 3
4	ADD R7, R1	0011 1001 0000	0x390	$R7 = 0 + 1 = 1$
5	ADD R7, R2	0011 1010 0000	0x3A0	$R7 = 1 + 2 = 3$
6	ADD R7, R3	0011 1011 0000	0x3B0	$R7 = 3 + 3 = 6$
7	JZR R0, 4	1100 0000 0100	0xC04	Jump to address 4

7. FLAG MAPPING ON BASYS3

Flag	LED	Pin	Meaning	When ON
Zero Flag	LD14	U14	ALU result = 0	Result is exactly 0
Overflow Flag	LD15	V14	Signed overflow	Result outside -8 to +7